

CE 310

Review Session

Week 8 · Session 1 · Connections Across Weeks 1–7

University of Arizona · Dr. Wooyoung Jung · Fall 2026

Today

- Walk through Weeks 1–7 one at a time: **core idea** → **worked number** → **common mistake**
- Nothing to turn in — ask questions any time
- Full detail (formula sheet, all worked examples) is in the **Study Guide** (Wk08.02) — this is the highlight reel, not a replacement
- Wed = Concept Exam (closed computer) · Fri = In-Class Exercise Practicum (open Excel)

Week 1 — The Flaw of Averages

Using the **average** to represent a whole distribution breaks down whenever the input–output relationship is nonlinear. For equipment sizing, the **peak** matters more than the mean.

Worked number. Tucson outdoor temp: $\mu = 68.75^{\circ}\text{F}$, $\text{MAX} \approx 106.9^{\circ}\text{F}$. A chiller sized for 68.75°F fails on every day above its rated capacity.

Common mistake: reporting only the mean and calling the system "adequate on average." Average-adequate $\neq$ peak-adequate — always check the tail.

Week 2 — Spread and Shape

STDEV, SKEW, and IQR describe how spread out and asymmetric a distribution is. Histograms/box plots reveal shape that a single number hides.

Worked number. Cooling_kWh in Tucson: $\sigma = 25.3$ kWh, positive skew (long right tail — a few peak-cooling hours dwarf the median).

Common mistake: STDEV (sample, $\div(n-1)$) vs. STDEVP (population, $\div n$). With engineering samples, use STDEV.

Week 3 — Conditional Probability

$P(A \mid B)$ = "given B happened, what's the chance A also happened?" = joint count $\div$ conditioning count.

Worked number. $P(\text{BusinessHours}=\text{"Yes"} \mid \text{Season}=\text{"Summer"}) = \text{COUNTIFS}(\text{Season}, \text{"Summer"}, \text{BH}, \text{"Yes"}) / \text{COUNTIF}(\text{Season}, \text{"Summer"})$.

Common mistake: dividing by the full dataset (8,760) instead of the conditioning count. $P(A|B)$ denominates on B, not on everything.

Week 3 — Reversing the Conditional (Bayes)

$P(A \mid B)$ and $P(B \mid A)$ are **different numbers** — same joint count, different denominators. Reversing one and quoting it as the other is the classic error.

Worked number. Detector fires on 90% of faulty units and 4% of healthy ones; 5% of units are faulty. $P(\text{Fault} \mid \text{Alarm}) = (0.90 \times 0.05) / (0.90 \times 0.05 + 0.04 \times 0.95) = 0.045 / 0.083 \approx \mathbf{54\%}$ — not 90%.

Common mistake: reporting the detector's hit rate as the answer. A low base rate keeps the posterior far below it — healthy units are 19× more numerous, so they generate most of the alarms.

Week 4 — Poisson Distribution

Models the count of discrete, independent events in a fixed interval, given rate λ .

Worked number. $\lambda = 3$ failures/month $\rightarrow P(X=0) = \text{POISSON.DIST}(0,3,\text{FALSE}) = e^{-3} \approx 0.0498$ ($\approx 5\%$ chance of zero failures).

Common mistake: using λ for the wrong time window. Rate = 6/hr but need a 15-min probability?
Convert first: $\lambda = 6/4 = 1.5$.

Week 4 — The Binomial

Fixed number of trials n , same success probability p each, want exactly k :

$P(X = k) = C(n, k) \cdot p^k \cdot (1-p)^{n-k}$, with $C(n, k) = n! / [k!(n-k)!]$.

Worked number. 8 valves, 15% each fail inspection $\rightarrow P(\text{exactly } 2) = C(8, 2)(0.15)^2(0.85)^6 = 28 \times 0.0225 \times 0.3771 \approx \mathbf{23.8\%}$.

Binomial or Poisson? Binomial needs a fixed n and per-trial p ("5 sensors, 10% each"). Poisson needs only a rate over an interval ("0.5 overflows/year").

Common mistake: dropping $C(n, k)$. Without it you get the probability of one *specific* arrangement, not of any k of the n .

Week 5 — Normal Distribution & Z-Scores

Normal = symmetric bell curve; $z = (x - \mu) / \sigma$ standardizes any Normal variable. 68–95–99.7 rule for $\pm 1\sigma/\pm 2\sigma/\pm 3\sigma$.

Worked number. Beta concrete: $\mu=4,367$ psi, $\sigma=484$ psi $\rightarrow P(f_c < 4,000) =$
`NORM.DIST(4000,4367,484,TRUE)` $\approx 22.4\%$. Beta fails ACI 318 (max 10% below spec). Alpha:
`NORM.DIST(4000,4632,344,TRUE)` $\approx 3.3\%$ — compliant.

Common mistake: using `NORM.DIST(...,FALSE)` — that's density, not probability. Use `TRUE` (cumulative) for a probability.

Week 6 — Empirical Percentiles & Exceedance

LARGE/PERCENTILE compute percentiles directly from ranked data — no distributional assumption needed. Exceedance = $P(X > \text{threshold})$.

Worked number. Traffic speed (500 rows): $\text{NORM.INV}(0.85, 54.4, 7.8) = 62.5$ mph (Normal model 85th %ile) vs. $\text{LARGE}(\text{Speed_mph}, 75)$ from ranked data (empirical 85th %ile; $15\% \times 500 = 75$). Models agree near the 85th percentile; gaps widen at extreme tails.

Common mistake: expecting $\text{PERCENTILE}(\text{data}, 0.85)$ and $\text{LARGE}(\text{data}, 75)$ to match exactly in a 500-row dataset. PERCENTILE interpolates; LARGE returns an actual data value — close, not identical.

Week 7 — Correlation and Covariance

Pearson r (–1 to 1) measures linear relationship strength/direction; r^2 = fraction of variance explained. Covariance has units, r doesn't.

Worked number. $r(\text{Outdoor_Temp_F}, \text{Cooling_kWh}) = 0.7752$ all-hours vs. 0.9539 business-hours-only — stronger during business hours because the HVAC system runs at full, predictable capacity.

Common mistake: reading $r \approx 0$ as "no relationship." It only rules out a LINEAR one — a strong U-shaped pattern can still have $r \approx 0$.

Questions?

Study Guide (wk08.02) has the full formula sheet and every worked example.

Wed: Concept Exam · Fri: In-Class Exercise Practicum

You've done this work seven times already. Trust it.